

EEE4114F

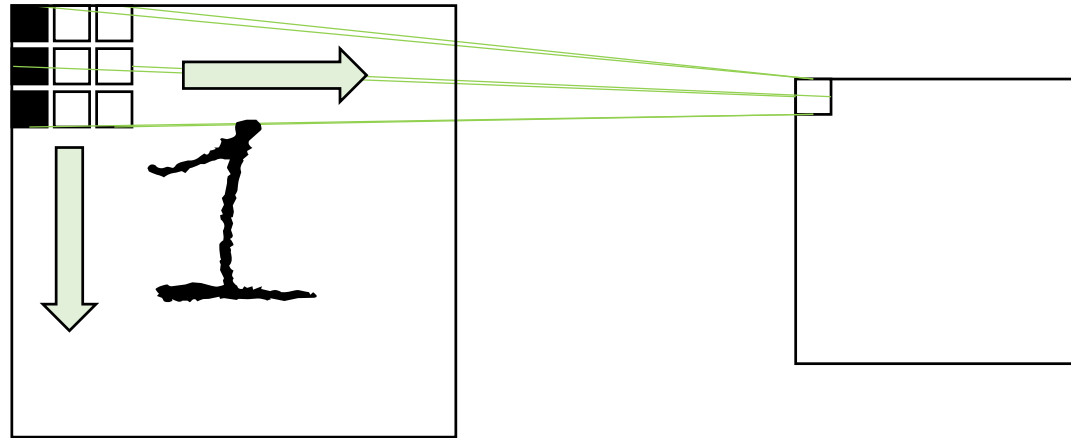
Parameter Sharing Details



Parameter sharing

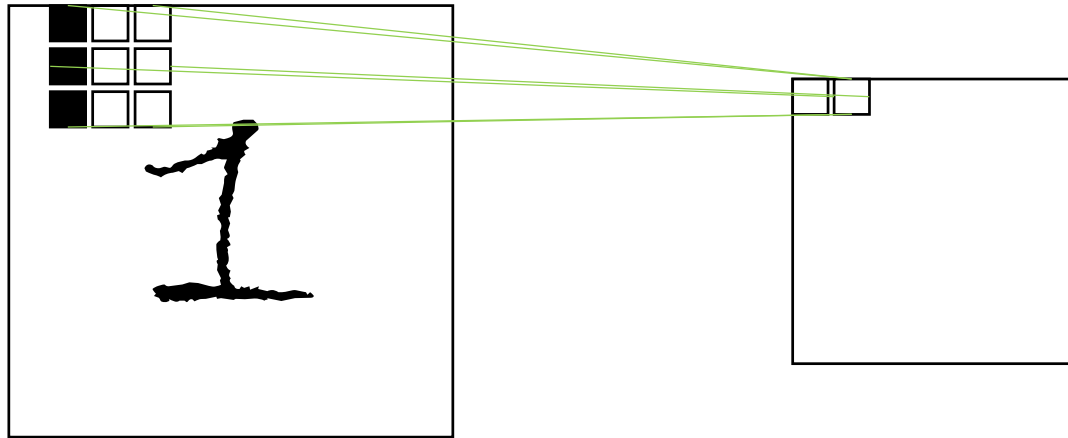
Slide the kernel around the input

$$z = \sum_{j=1}^9 w_j x_j, a = \sigma(z)$$



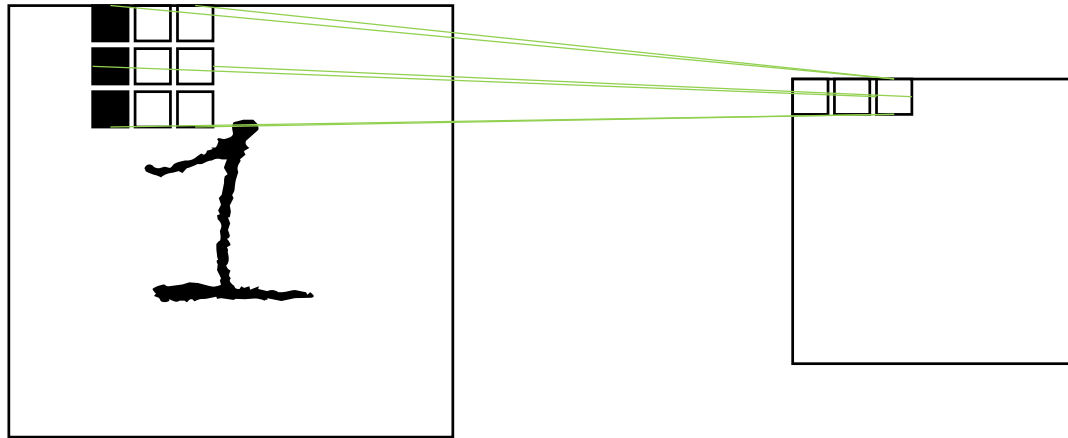
Parameter sharing

$$z = \sum_{j=1}^9 w_j x_j$$



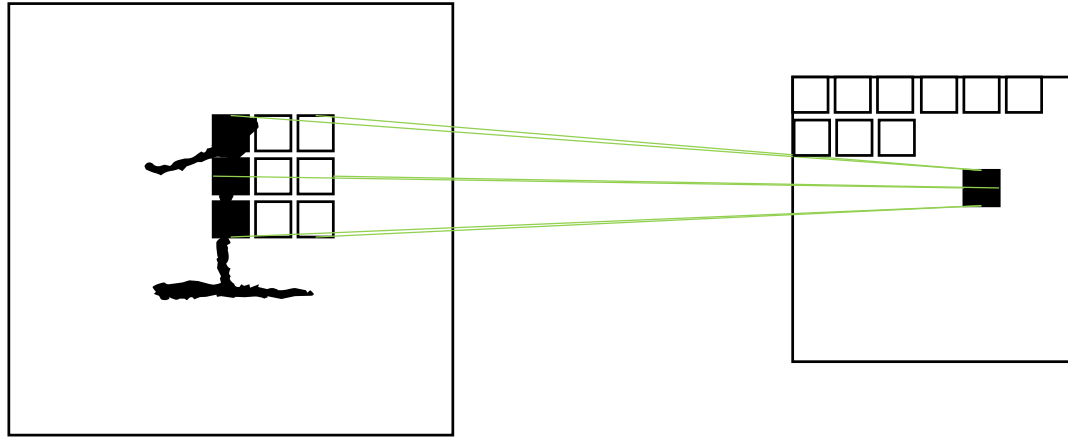
Parameter sharing

$$z = \sum_{j=1}^9 w_j x_j$$

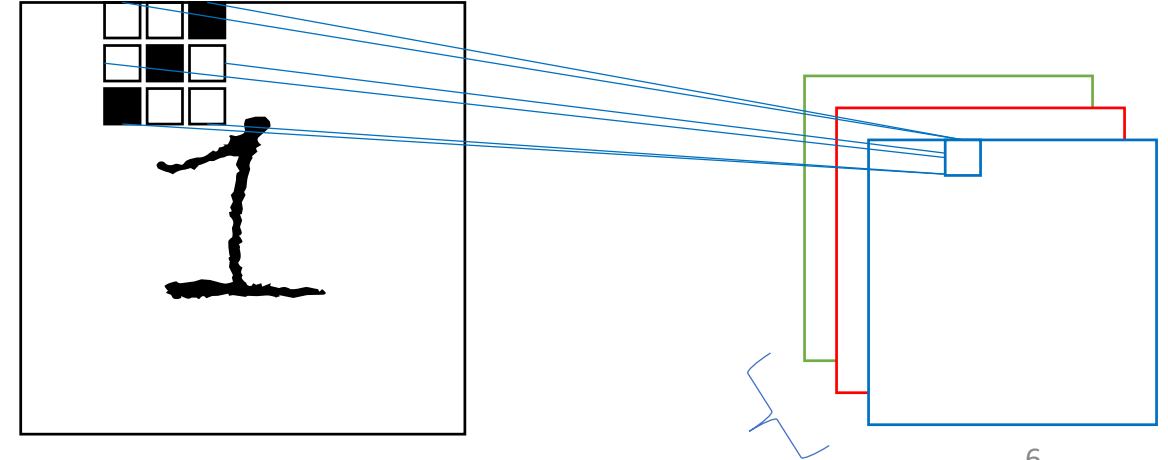
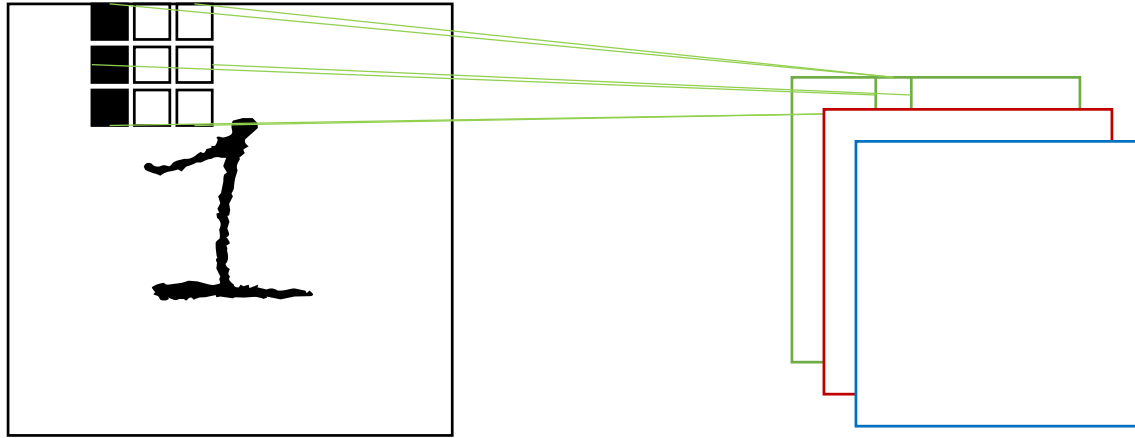


Parameter sharing

$$z = \sum_{j=1}^9 w_j x_j$$

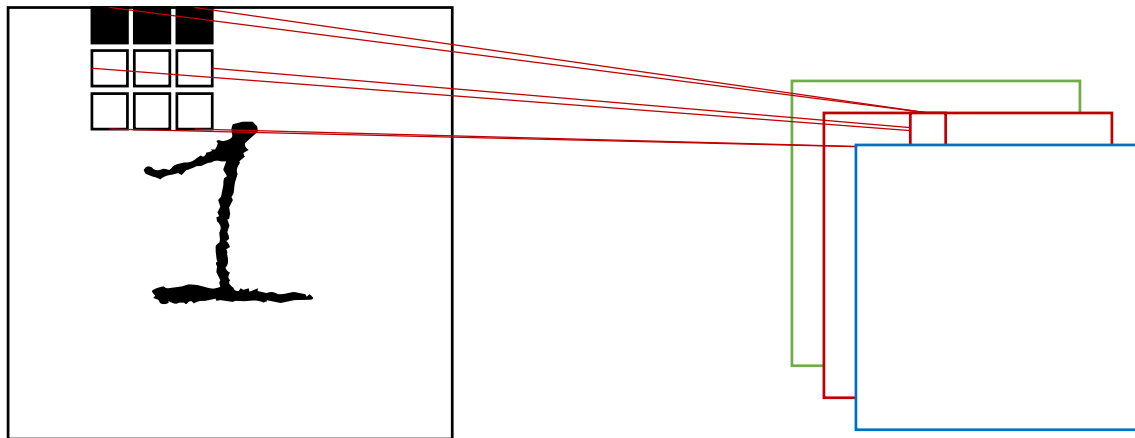


Multiple kernels

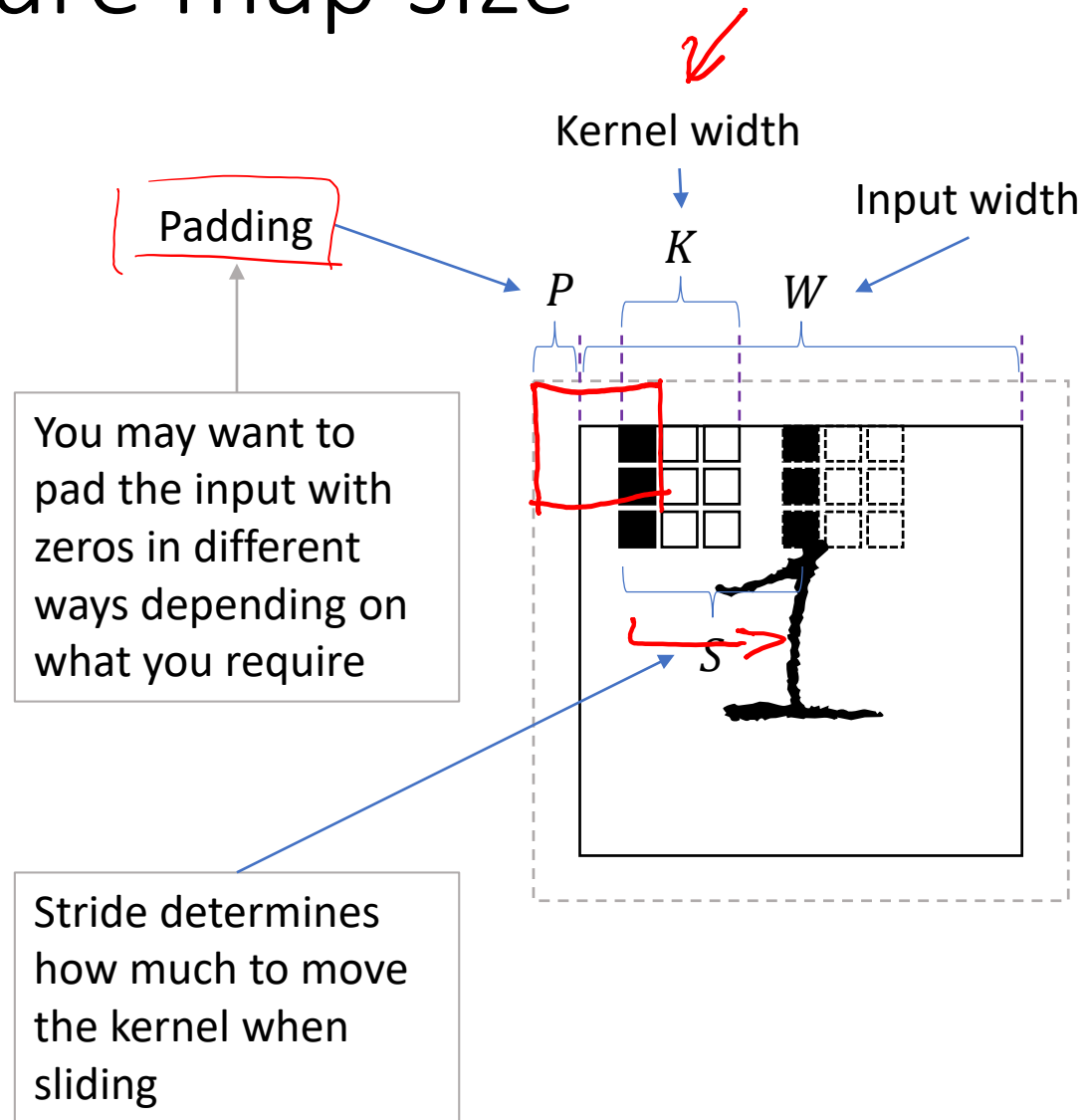


3 kernels = 3 feature maps

6

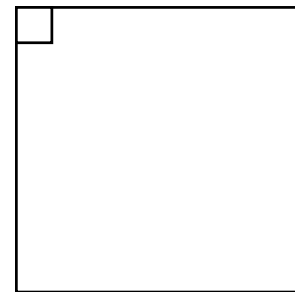


Feature map size

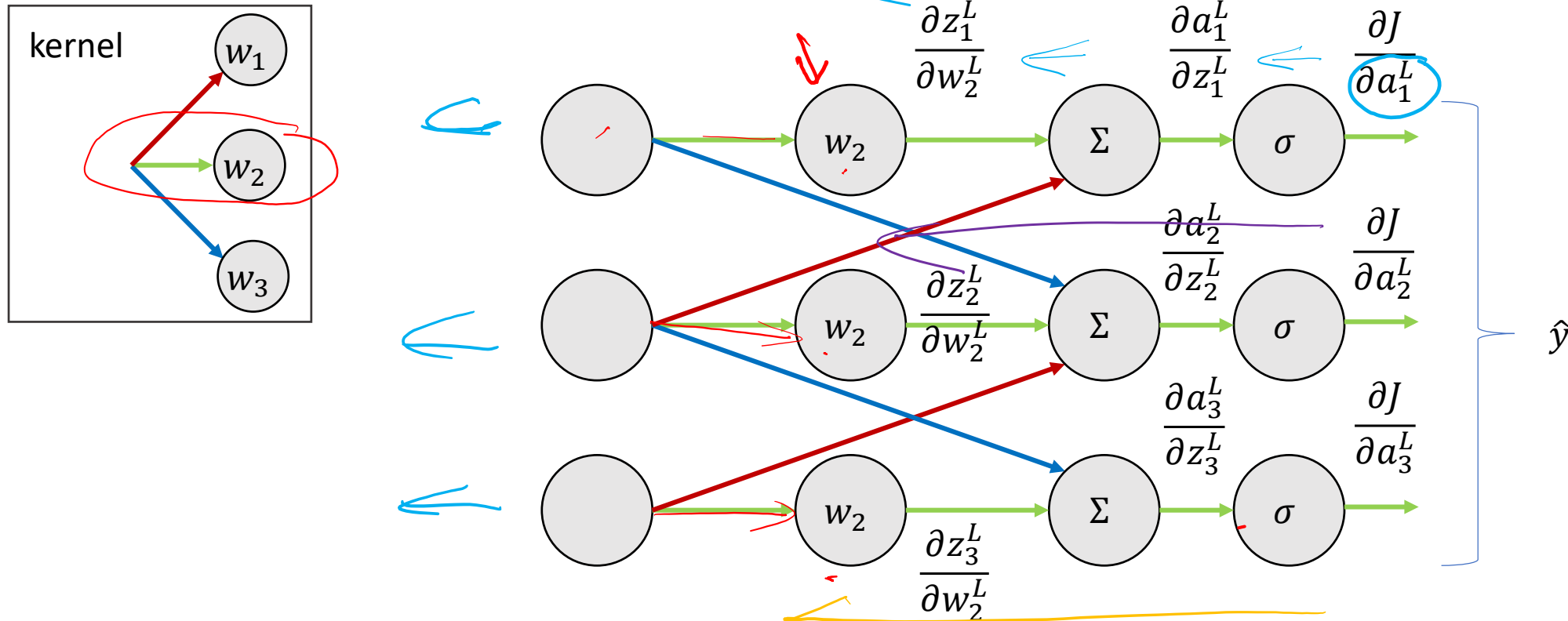


$\lfloor \rfloor \rightarrow \text{floor}$

$$O = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1$$



Backpropagation with Shared Parameters



$$\frac{\partial J}{\partial w_2^L} = \frac{\partial J}{\partial a_1^L} \cdot \frac{\partial a_1^L}{\partial z_1^L} \cdot \frac{\partial z_1^L}{\partial w_2^L} + \frac{\partial J}{\partial a_2^L} \cdot \frac{\partial a_2^L}{\partial z_2^L} \cdot \frac{\partial z_2^L}{\partial w_2^L} + \frac{\partial J}{\partial a_3^L} \cdot \frac{\partial a_3^L}{\partial z_3^L} \cdot \frac{\partial z_3^L}{\partial w_2^L}$$



Next up:

Padding

